

Enhancing Academic Literature Reviews with a Retrieval Augmented Generation (RAG)
System

By

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Abstract

The Technology Innovation Management (TIM) program seeks to explore new advancements, such as Retrieval-Augmented Generation (RAG), to enhance the efficiency and effectiveness of academic literature reviews. This project addresses this goal by developing a custom RAG system specifically designed for the TIM domain. The client for this project is the TIM program at Carleton University, led by Professor Tony Bailetti. Key deliverables include a modular retrieval and summarization pipeline, pathway mapping & categorization, and AI generated augmented queries to streamline academic research tasks. To achieve these results, the project uses a design science methodology, combining iterative prototyping, user feedback, and evaluation metrics to ensure the system's alignment with TIM research needs. This project contributes to enhancing the efficiency and reliability of literature review processes with the help of generative artificial intelligence, empowering researchers with innovative tools, and positioning the TIM program as a leader in using advanced AI for academic research.

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Glossary of Terms

Term	Definition	Source
Answer Faithfulness	A measure of whether the generated output of a RAG system remains grounded in and consistent with the retrieved information.	Saad-Falcon et al. (2024)
Augmentation	The process in RAG systems of combining retrieved external data with user input to create a richer context for generating accurate and relevant responses.	Gao et al. (2024)
Context Relevance	A metric evaluating how well retrieved information aligns with the input query in RAG systems, ensuring the appropriateness of sources for generating responses.	Saad-Falcon et al. (2024)
Design Science (DS)	A research methodology focusing on the creation and testing of innovative solutions, blending creative design with explanatory science to address complex challenges and contribute to both theory and practice.	Romme (2023)
Evaluation Frameworks	Systems or methods designed to assess the performance of RAG systems across metrics like relevance, faithfulness, and accuracy without relying entirely on human annotations.	Es et. al. (2023)
Hallucination	The generation of incorrect or non-factual information by AI systems, a challenge addressed by integrating retrieval mechanisms in RAG systems.	Gao et al. (2024)
Large Language Models (LLMs)	Advanced AI systems trained on vast datasets to generate human-like text, often used in natural language processing tasks.	Gao et al. (2024)
Query Template	Predefined or customizable structures used to optimize information retrieval in RAG systems, tailored to specific tasks or domains.	Han et al. (2024)
Retrieval-Augmented Generation (RAG)	A framework that combines large language models with real-time information retrieval to enhance factual accuracy and reliability by grounding responses in dynamically retrieved content.	Gao et al. (2024)
Systematic Literature Review (SLR)	A structured approach to identifying, evaluating, and synthesizing research to address specific research questions, often involving stages like literature search, screening, data extraction, and synthesis.	Han et al. (2024)

1 Introduction

The Technology Innovation Management (TIM) program at Carleton University is committed to exploring new advancements to enhance research practices. This project specifically examines the potential of Retrieval-Augmented Generation (RAG) systems to streamline academic literature reviews. RAG systems have emerged as a transformative technology by combining the knowledge-generation capabilities of large language models (LLMs) with real-time information retrieval from external sources, thereby addressing issues such as hallucinations and the reliance on static pre-trained knowledge bases (Gao et al., 2024). While current literature highlights the advantages of RAG in improving factual accuracy and relevance, limited research exists on domain-specific customization, particularly for academic literature reviews such as those sought by TIM faculty and students (Han et al., 2024).

Existing solutions for academic literature reviews often lack the ability to integrate domain-specific knowledge effectively, a gap that this project aims to address by tailoring RAG systems for the TIM domain. The system will not only enhance retrieval accuracy but also ensure outputs are traceable and trusted by researchers. While studies such as those by Es et al. (2024) and Saad-Falcon et al. (2024) propose evaluation frameworks for RAG systems, they emphasize general applications, leaving the need for domain-specific adaptations unfulfilled. This project will adapt the insights from these studies, focusing on developing a dynamic query pathway mapping module and AI-generated augmented queries, tailored to the research needs of TIM students and faculty. Theoretical advancements in RAG systems are grounded in their ability to combine retrieval, augmentation, and generation processes effectively (Han et al., 2024;

Gao et al., 2024). Retrieval focuses on identifying the most relevant documents, augmentation involves contextualizing the retrieved information, and generation integrates this data to produce coherent outputs. However, these processes present challenges such as low precision in retrieval and hallucinated outputs in generation, particularly when applied to niche fields (Han et al., 2024). By leveraging modular pipeline designs, this project aims to address these limitations and enhance the specificity and reliability of RAG outputs.

1.1 Objective

The objective of this project is to design and develop an RAG system that delivers relevant, traceable, and trusted outputs for academic literature reviews. The system will focus on enhancing output quality while addressing challenges like hallucination, lack of citations, and domain-specific adaptation.

1.2 Deliverables

The key deliverables of this project include:

- (1) A Modular Retrieval and Summarization Pipeline. Found in section 4.1.
- (2) Pathway Mapping and Categorization. Found in section 4.2.
- (3) AI Generated Augmented Queries. Found in section 4.3.
- (4) Synthesis of insights and lessons learned. Found in section 4.4.

1.3 Relevance

The project is highly relevant to academic researchers and practitioners who rely on systematic literature reviews to synthesize evidence. By ensuring traceable and high-quality outputs, the proposed RAG system addresses the critical needs of accuracy,

efficiency, and trustworthiness in literature synthesis. The project also aligns with emerging trends in AI-enhanced tools, offering a solution adaptable to various research domains.

1.4 Contribution

This project makes practical and research contributions focusing on enhancing the efficiency, accuracy, and usability of Retrieval-Augmented Generation (RAG) systems for academic literature reviews. It provides insights into the application of RAG systems by analyzing existing approaches and identifying key challenges related to trust, relevance, and traceability, it highlights practical ways to improve literature synthesis. The pathway mapping and categorization module semantically classifies queries into TIM research domains, offering a practical tool for contextualizing literature reviews and aligning them with thematic pathways. From a practical perspective, this project develops a modular retrieval and summarization pipeline that simplifies the literature review process, reducing the manual workload for researchers while improving the quality and traceability of results. The augmented queries enable the dynamic generation of diverse, context-rich research prompts, improving the coverage and precision of literature retrieval. Additionally, the lessons learned, and insights offer actionable guidance for researchers and practitioners seeking to integrate RAG systems into their workflows.

1.5 Overview of method

This study employs a systematic approach to develop a Retrieval-Augmented Generation (RAG) system tailored for TIM reviews. First, a systematic literature review was conducted to identify relevant metrics, concepts, and principles to guide the RAG

system design. This step established the foundation for improving trust, relevance, and traceability in RAG outputs.

Next, a modular RAG pipeline will be developed, integrating domain-specific knowledge categories and retrieval components optimized for TIM research tasks. A semantic pathway mapping module will be used to group queries by domain relevance, improving retrieval contextualization. To support researchers, AI-driven query augmentation techniques will be applied to produce diverse and focused search prompts. The system will be tested iteratively, incorporating feedback from TIM researchers and academic stakeholders to ensure usability and relevance. Insights gained during testing will inform further refinements to align the system with user needs. Finally, lessons learned throughout the process will be documented to highlight practical challenges, achievements, and areas for future improvement.

1.6 Organization of the document

This project is organized into six chapters. Chapter 2, "Literature review," explores the literature on RAG systems and their relevance to academic research. Chapter 3, "Research method," details the design science approach and the specific methods used to achieve the stated objectives. Chapter 4, "Results," presents the developed RAG system and its evaluation results, showcasing its functionality and performance. Chapter 5, "Discussion of results," analyzes the findings, offering insights and exploring the broader implications of the results. Finally, Chapter 6, "Conclusions, limitations, and future research," summarizes the project's contributions, acknowledges its limitations, and outlines potential directions for future research.

2 Literature Review

The rapid integration of Artificial Intelligence (AI) into academic research processes has changed the ways in which knowledge is synthesized and applied. This review aims to support the development of a Retrieval-Augmented Generation (RAG) system tailored for academic literature reviews, ensuring that outputs are relevant, traceable, and trusted by users. By synthesizing insights from four key streams: (1) Addressing Trust Issues in RAG Systems, (2) Design Science Methodology, (3) Systematic Literature Review, and (4) The Role of AI in Academic Literature. The following four sections review each stream, and the final fifth section is a summary and synthesis of the key lessons from the literature, across all four streams, that are most relevant to this research.

The review is guided by research questions formulated using the PCC Framework (Population, Concept, and Context). This framework ensures clarity and precision in the development of research questions, which were subsequently used to search for articles on ELICIT AI and Google Scholar. The research questions are presented in the table below

Table 1: Review Questions

Stream	Research question
Addressing Trust Issues in RAG Systems	• What strategies can be implemented to ensure the relevance and traceability of RAG-generated outputs? • What are the key requirements for ensuring trust, relevance, and traceability in RAG systems? • What measurable outputs can be evaluated from existing RAG systems?
Design Science Methodology	• How can the design science methodology be used to create an LLM based application prototype?

Systematic Literature Review	• What are the features of the best systematic literature reviews? • How can systematic literature review practices be enhanced using RAG systems?
The role of AI in Academic Literature	• How does integrating AI in literature reviews influence the efficiency and relevance of academic research? • What are the benefits and limitations of current AI tools in addressing domain-specific literature review needs?

To maintain methodological rigor, the review includes peer-reviewed articles published between 2018 and 2024, with abstracts screened to ensure relevance. Starting with an initial pool of 100 articles, duplicates and irrelevant documents were excluded, resulting in 62 articles that informed the synthesis. The search process also used LitMAP to visualize connections between articles in each stream. These visualizations highlight interconnections between studies, although in the Design Science Methodology stream, no links were observed, indicating the independence of its contributing papers.

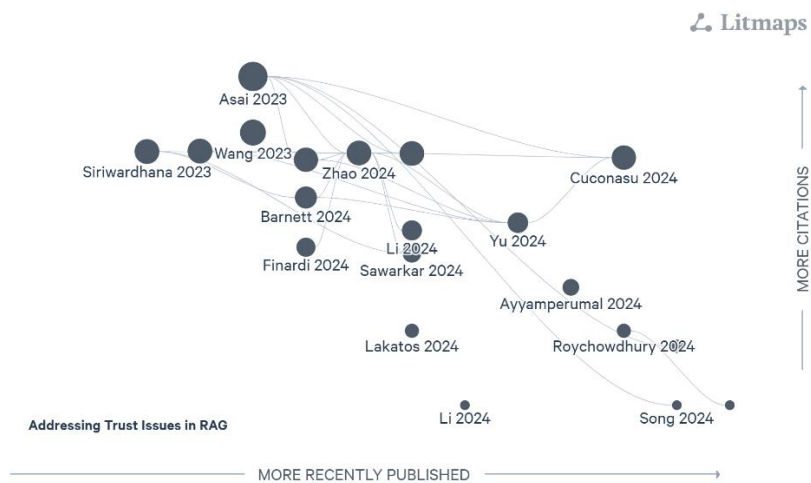


Figure 1: Trust, Relevance, and Traceability in RAG Systems

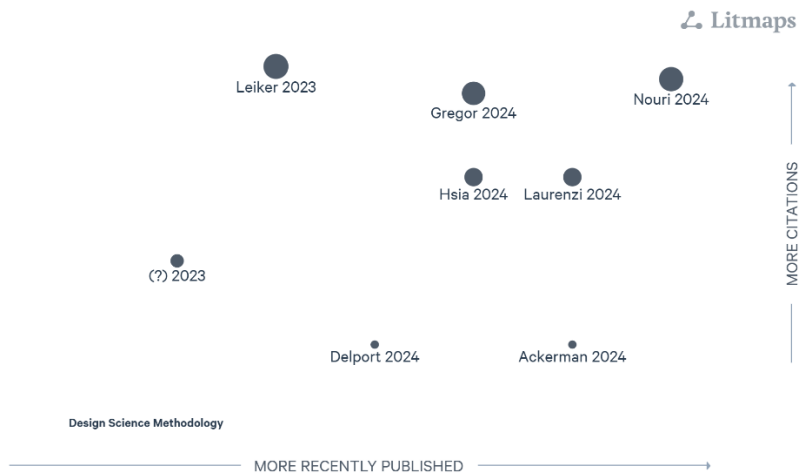


Figure 2: Design Science Approaches for RAG System Development

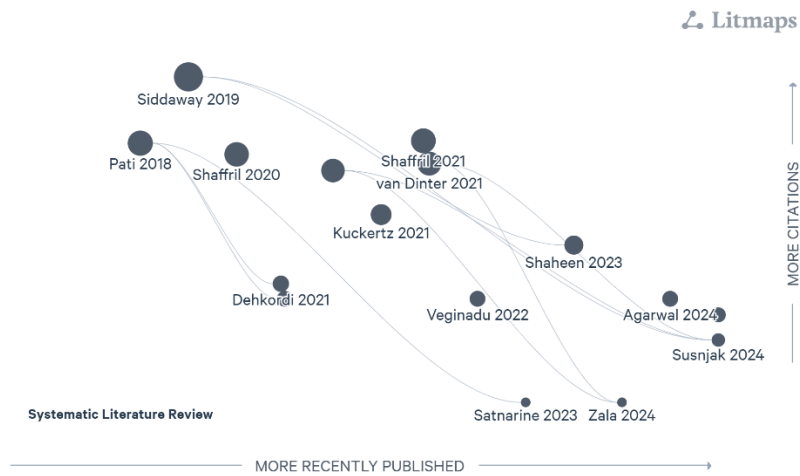


Figure 3: Methodological Frameworks for Systematic Literature Reviews

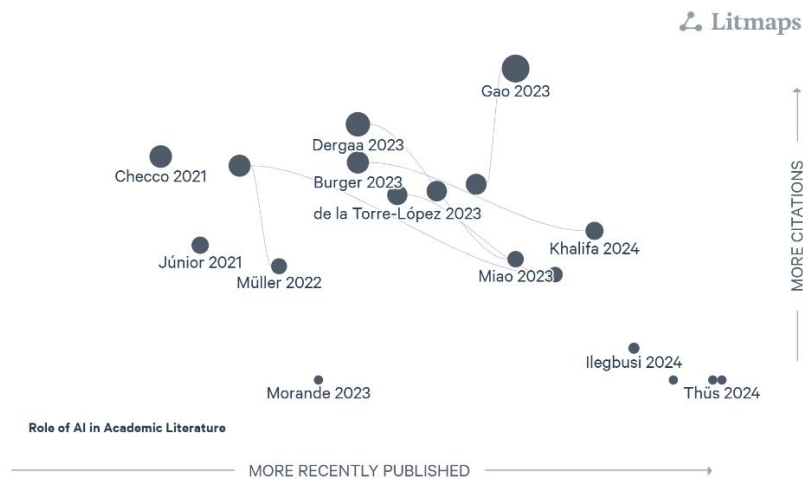


Figure 4: AI Tools and Their Role in Academic Literature

2.1 Addressing Trust Issues in RAG Systems

Ensuring trust, relevance, and traceability in Retrieval-Augmented Generation (RAG) systems is critical for their deployment in academic literature tasks. The ability to ground

generated outputs in external sources improves system reliability by reducing hallucinations and ensuring that the retrieved knowledge remains up to date (Zhao et al., 2024). Query-based augmentation, such as integrating retrieval methods like DPR or BM25 with input queries, directly enhances relevance while grounding outputs in reliable sources (Zhao et al., 2024). For domain-specific tasks, retrieval approaches require modality-specific adjustments (e.g., for text, image, or code), ensuring alignment with the target domain while maintaining traceability (Zhao et al., 2024).

Evaluation frameworks like TRUST-SCORE and GRAMMAR have emerged to systematically measure traceability across multiple dimensions. TRUST-SCORE evaluates outputs based on grounded refusals, response truthfulness, and attribution groundedness, ensuring that outputs align with referenced documents or refuse to answer when sources are insufficient (Song et al., 2024). The GRAMMAR framework isolates robustness issues such as query ambiguity and retrieval gaps, which undermine traceability and trustworthiness in generated outputs (Li et al., 2024).

To improve trust in academic RAG systems, strategies like Trust-Align have been developed. Trust-Align enhances traceability by aligning LLMs using a grounded dataset of labeled queries, documents, and responses. Empirical benchmarks demonstrate improvements of 16-22% in grounded refusals and 26-31% in citation accuracy, underscoring its value in addressing hallucination and improper citation challenges (Song et al., 2024). Further, tools like RAFT (Retrieval-Augmented Fine-Tuning) focus on minimizing distractions from irrelevant documents while encouraging detailed, citation-supported reasoning, leading to improved trust and traceability (Zhang et al., 2024).

Metrics like faithfulness, factual correctness, and answer relevance provide measurable benchmarks to quantify trust and traceability in RAG outputs (Roychowdhury et al., 2024). Faithfulness assesses whether generated content aligns with retrieved contexts, while answer relevance evaluates how well outputs address input queries. Tools such as RAGAS refine these metrics to provide interpretable intermediate results, enhancing traceability evaluation in domain-specific contexts (Roychowdhury et al., 2024).

Despite these advancements, significant challenges remain. Internal knowledge within RAG systems can conflict with external data, undermining trust in outputs (Zhou et al., 2024). Systems must prioritize accurate and reliable sources to generate verifiable and traceable content. Implementing interactive feedback mechanisms allows users to validate citations and flag inaccuracies, improving user trust and system relevance (Venkit et al., 2024).

Additionally, strategies such as hybrid retrieval (combining semantic search and sparse/dense retrieval) enhance precision and ensure that outputs are grounded in the most relevant documents (Sawarkar et al., 2024). Contextual filtering and re-ranking processes refine retrieved knowledge before generation, further improving relevance and traceability (Li et al., 2024).

In high-stakes applications, frameworks such as SELF-RAG introduce a reflection mechanism where models critique their outputs, identifying whether retrieval is necessary and evaluating retrieval accuracy. By validating outputs at each generation step, SELF-RAG enhances traceability and trust (Asai et al., 2023). Metrics like relative maximum scores quantify the gap between baseline and optimal retrieval performance, offering additional precision in system evaluation (Finardi et al., 2024).

2.2 Features of a RAG LLM

The table below presents the selected candidate features to be implemented in the prototype, distilled from an initial set of 28 features identified in the literature. Each feature was evaluated and prioritized based on its feasibility and relevance to the project's objectives. The five selected features represent practical and impactful strategies to address the inefficiencies of current RAG systems and lay a foundation for a domain-specific solution for academic literature reviews. The complete list of the 28 features can be found in the appendix.

Table 2: Selected Features

Feature	Category	Description	Relevance
Semantic Filtering	Filtering Techniques	Removes irrelevant documents based on semantic matching with the query.	Improves content relevance by filtering out noise.
Query Augmentation	Retrieval Optimization	Expands user queries to capture diverse or more precise perspectives.	Facilitates the discovery of implicit knowledge and transformative insights.
Metrics for Evaluation	Evaluation Techniques	Uses Relevance, Groundedness, and Retrieval Relevance to evaluate content alignment, factual support, and retrieval accuracy.	Essential for evaluating prototype effectiveness, Provides quantitative insights into system reliability and relevance.
Modular RAG Framework	Architecture	Decouples retrieval, augmentation, and generation processes for adaptability.	Increases flexibility and scalability of the RAG system.
Static Query Interface	User Interface	Provides a user interface where users can submit review queries and receive generated outputs.	Offers a usable front-end for query input and result display.

2.3 Design Science Methodology

The adoption of Design Science Methodology (DSM) offers a structured, iterative approach for developing robust systems that meet specific domain requirements.

Design science principles have been widely applied to information systems and are

adaptable for Retrieval-Augmented Generation (RAG) systems, particularly in academic and domain-specific tasks (Kumar, 2024).

2.3.1 Phases of the Design Science Process

Several studies outline systematic design phases that align with the development lifecycle of RAG systems. Kumar (2024) introduces a five-phase methodology starting with problem identification, which defines the scope and core challenges of the system. The next phase, requirements definition, involves gathering functional and non-functional requirements through interviews, literature reviews, and motivational analysis. In the design and development phase, system architecture is formalized using structural diagrams, behavioral diagrams, and deployment views to create a robust framework. This is followed by evaluation, where functional and usability testing ensures that the design goals are met. Finally, the communication of results phase focuses on disseminating outcomes to ensure transparency and facilitate future adaptability. These iterative phases emphasize expert validation, a critical strategy for enhancing the relevance, accuracy, and traceability of RAG systems.

In addressing complexity, combinatorial optimization frameworks can be employed to evaluate system parameters systematically. Ackerman et al. (2024) propose a three-stage solution design process that includes exploration, analysis, and control phases. Combinatorial techniques reduce design space complexity, limiting evaluations to significant parameter interactions and ensuring efficient alignment of inputs, prompts, and outputs.

2.3.2 Human-in-the-Loop and Iterative Cycles

To ensure accuracy and trustworthiness, design science approaches incorporate a human-in-the-loop process. Leiker et al. (2023) adopt this strategy to validate outputs and ensure alignment with quality standards, achieving 86.3% accuracy and 88.0% clarity. Such iterative frameworks combine user input with automated systems, ensuring that generated outputs meet domain-specific requirements. Additionally, Nouri et al. (2024) highlights the importance of task decomposition, where complex tasks are divided into manageable subtasks to improve scalability and reliability in LLM-driven systems.

The design science methodology also includes rigorous testing cycles, such as alpha-testing (feasibility and robustness) and beta-testing (reliability and generalizability) to enhance system performance (Romme, 2023). These iterative evaluations ensure systems remain adaptable and trustworthy when applied to academic RAG contexts. The following sections highlight the design requirements and principles that guide system development, ensuring alignment with context-specific needs and iterative validation processes (Gregor & Hevner, 2024).

2.3.3 Design Requirements

(i) Iterative Development and Validation

The development of RAG systems should involve iterative cycles of refinement informed by user feedback. These cycles ensure that the system's outputs align with academic standards and address dynamic user needs effectively. This approach enhances both

the traceability and relevance of the system's output by continuously validating its performance against stakeholder expectations.

(ii) Framework-Based Design

The use of structured methodologies, such as the Design Science Research Methodology (DSRM), ensures rigor and replicability in the development process. A framework-based design provides a systematic roadmap for building and evaluating RAG systems, particularly those tailored for literature reviews.

(iii) Trustworthiness and Relevance Metrics

The system must integrate clear, measurable criteria for evaluating trustworthiness and relevance. This includes implementing real-time retrieval quality assessments and leveraging user-centric feedback mechanisms to ensure output meets scholarly expectations.

(iv) Domain-Specific Adaptation

To cater to the unique demands of academic literature reviews, the system should incorporate modular components tailored to specific domains. This adaptation ensures that the system provides outputs that are both relevant and contextually accurate.

(v) Human-in-the-Loop Validation

Expert validation is essential at critical stages of development. This ensures that the artifacts not only meet theoretical benchmarks but also align with practical applications, enhancing user trust and confidence in the system.

2.3.4 Design Principles

(i) User-Centric Iteration

Feedback loops should be an integral part of the development process, enabling continuous improvement based on user input and system performance metrics. This principle ensures that the system evolves in alignment with user needs.

(ii) Integration of Theory and Practice

The system's design should bridge theoretical foundations with practical applications. By aligning theoretical models with real-world use cases, the RAG system can achieve both academic rigor and practical viability.

(iii) Motivation-Driven Development

The system's design should be informed by motivational factors, including the specific needs of researchers engaged in literature reviews. This ensures that the system remains aligned with user priorities and research tasks.

(iv) Test-First and Benchmarking Strategy

Incorporating a test-first approach, alongside predefined benchmarks, ensures early validation of the system's functionality and adaptability. This principle emphasizes the importance of continuous evaluation to align the system with its intended objectives.

2.4 Systematic Literature Review (SLR)

Systematic literature reviews (SLRs) are distinguished by their clear and well-justified research questions. These questions may be broad, addressing state-of-the-art developments in a field, or specific, evaluating empirical evidence on a particular

phenomenon. A well-defined research question not only ensures focus but also justifies the review's value to the research community (Kuckertz & Block, 2021). Transparency in inclusion and exclusion criteria is equally critical. Stating these criteria explicitly and justifying them ensures that decisions about study selection are reproducible and bias-free (Kuckertz & Block, 2021; Siddaway et al., 2019).

Acknowledging prior reviews and identifying gaps or limitations is another hallmark of high-quality SLRs. This ensures the relevance of the current review and establishes its contribution to the field (Kuckertz & Block, 2021). Methodological rigor is achieved through structured frameworks such as PRISMA and Cochrane guidelines, which emphasize systematic searching, critical analysis, and replicability (Pati & Lorusso, 2018; Shaheen et al., 2023). A structured process including scope definition, eligibility criteria, literature searching, data extraction, quality assessment, and result synthesis ensures consistency and minimizes bias (Veginadu et al., 2022).

The integration of AI tools offers promising solutions for enhancing SLR workflows. Tools like ChatGPT assist in refining research questions, while ontology-based systems like LitSonar improve search precision through automated query construction (Fabiano et al., 2024; Zala et al., 2024). Machine learning applications, such as ASReview, prioritize study screening to reduce manual workloads without sacrificing precision and recall (Fabiano et al., 2024). Natural language processing (NLP)-based systems facilitate accurate data extraction by automating the identification of objectives, methods, and outcomes with reduced error rates (Zala et al., 2024). AI-driven tools also streamline the synthesis of findings, consolidating results efficiently, though human

oversight remains essential to ensure contextual relevance and accuracy (Susnjak et al., 2024).

Fine-tuning large language models (LLMs) using domain-specific datasets further enhances automation in SLRs. Tools like LitLLM employ Retrieval-Augmented Generation (RAG) systems to ground outputs in retrieved literature, reducing hallucinations and improving factual accuracy (Agarwal et al., 2024). These systems also integrate text mining techniques and combinatorial frameworks to optimize the selection and synthesis of data (van Dinter et al., 2021). While automation significantly reduces the manual effort involved in traditional SLRs, balancing methodological rigor with efficiency remains a key challenge (Veginadu et al., 2022).

Human-AI collaboration plays a critical role in ensuring high-quality SLRs. While AI tools excel in automating repetitive tasks, human reviewers provide validation and interpretive insights that preserve the review's methodological integrity and contextual relevance (Zala et al., 2024). Finally, frameworks like PICO and SPIDER enhance precision in formulating research questions and identifying keywords, ensuring relevance and focus in systematic reviews (Shaffril et al., 2020).

2.5 Role of AI in Academic Literature

AI technologies, including machine learning (ML) and natural language processing (NLP), have transformed key academic tasks by improving efficiency and precision. For example, in peer review processes, AI automates repetitive tasks such as plagiarism detection, format checks, and reviewer-paper matching. This reduces workload and improves consistency while achieving outcomes comparable to human decisions

(Checco et al., 2021). However, AI systems cannot replace the domain expertise required for critical evaluation and intellectual judgment, which are essential for high-quality peer review (Checco et al., 2021).

In academic writing and content generation, generative AI tools like ChatGPT assist researchers by drafting manuscripts, summarizing papers, and generating research questions. These tools save time and reduce the cognitive effort associated with academic tasks (Dergaa et al., 2023). Despite these advantages, AI-generated content often lacks precision, includes fictitious references, and struggles with complex mathematical or technical details, raising concerns about the reliability and originality of such outputs (Dergaa et al., 2023).

AI has also advanced the automation of literature reviews by streamlining tasks such as search, screening, data extraction, and synthesis. Tools like LitSonar and LitLLM optimize search queries, prioritize relevant studies using machine learning, and produce concise summaries of findings. Retrieval-Augmented Generation (RAG) models have proven particularly effective by grounding outputs in reliable external sources, significantly reducing errors such as hallucinations and improving the traceability of results (da Silva Júnior & Dutra, 2021; Agarwal et al., 2024).

2.5.1 Ethical Considerations

Despite its benefits, AI introduces ethical challenges that cannot be overlooked. Plagiarism risks are heightened as AI tools can generate large volumes of text, potentially leading to improper attribution. Ensuring transparency in AI usage and adopting plagiarism detection tools like GPT-Zero are essential to maintaining academic

integrity (Miao et al., 2024). Additionally, AI models can reflect biases from their training data and occasionally produce hallucinated outputs—plausible yet incorrect information. Human oversight remains critical for validating outputs, ensuring they align with established academic standards (Gupta et al., 2024).

2.5.2 Efficiency and Relevance in Literature Reviews

AI has significantly improved the efficiency and relevance of literature reviews by automating traditionally labor-intensive tasks. Tools powered by ML and NLP, such as LitLLM and RAG systems, streamline the retrieval of domain-specific research by grounding outputs in reliable, external sources. This reduces noise in search results and enhances the precision of retrieved content (Gao et al., 2024).

RAG systems, in particular, address some limitations of traditional AI tools by integrating real-time retrieval capabilities with generative outputs. They ensure factual grounding, align outputs with relevant academic sources, and allow for modular adjustments to meet domain-specific requirements (Saad-Falcon et al., 2024).minimizing hallucinations and ensuring content aligns with verified references (Gao et al., 2024).

2.6 Novelty

This project introduces a novel approach to automating systematic literature reviews by developing a domain-aware RAG pipeline tailored to the Technology Innovation Management (TIM) context. While existing RAG systems primarily address general academic purposes, this project makes the following contributions:

- (i) TIM-Specific Pathway Mapping: A distinctive feature of this system is the pathway mapping module, which semantically classifies queries into

- predefined TIM research streams. This enhancement ensures better contextual relevance and supports more targeted, domain-specific reviews.
- (ii) **AI-Driven Query Augmentation Using PCC:** The system employs GPT-based prompt engineering structured around the Population-Concept-Context (PCC) framework. This structured augmentation improves the breadth of document retrieval, offering an innovative application of PCC in automated academic workflows.
 - (iii) **Integrated Evaluation via LLM Judge:** This project incorporates an LLM-based evaluation in addition to quantitative metrics. This dual-mode evaluation assesses the factual accuracy, coherence, and completeness of the generated outputs, ensuring a more nuanced and traceable performance assessment.

2.6.2 Limitations

Despite these strengths, challenges remain. Computational overhead, retrieval precision issues, and noisy datasets can hinder performance. Addressing these limitations requires careful fine-tuning of retriever-reader alignments and robust filtering mechanisms to enhance scalability and accuracy (Gao et al., 2024).

AI tools, while transformative, are not without drawbacks. Their reliance on training data means they may fail to adapt to niche or emerging research topics (Wang et al., 2024). Moreover, the quality of output still depends heavily on human validation to address errors, biases, and contextual misinterpretations (Miao et al., 2024; Morande, 2024).

Additionally, high computational demands can make some AI tools inaccessible to resource-constrained researchers (Gupta et al., 2024; Gao et al., 2024).

To mitigate these limitations, a collaborative approach is essential. Combining AI's speed and efficiency with human expertise ensures that outputs are not only accurate but also contextually relevant and reliable (Zala et al., 2024; Wagner et al., 2022). This balance between automation and oversight is critical for maintaining the integrity of academic literature reviews (de la Torre-López et al., 2023).

2.7 Summary and Synthesis of Key Findings

Table 3: Summary of Key Findings

Research Stream	Requirements for Trust, Relevance, Traceability	Current Measurable outputs	Sources
Trust in RAG Systems	- Trust: Mitigation of bias, transparency - Relevance: Accurate, domain-relevant retrieval - Traceability: Cited references, reproducible outputs	(1) Faithfulness (answer grounding) (2) Citation accuracy (3) Grounded refusals (4) Noise robustness	Lakatos et al. (2024) Li et al. (2024) Roychowdhury et al. (2024) Song et al. (2024) Zhao et al. (2024)
Design Science Methodology	- Trust: Iterative evaluation, human validation - Relevance: Domain-specific prompt tuning, accurate retrieval - Traceability: Faithful outputs, modular architecture for transparency	(1) Output evaluation frameworks (2) Usability testing (3) Retriever-reader alignment metrics	Gregor & Hevner (2024) Hsia et al. (2024) Kumar (2024) Laurenzi et al. (2024) Nouri et al. (2024)
Systematic Literature Review	- Relevance: Comprehensive, accurate retrieval - Traceability: PRISMA-based outputs, source clarity - Trust: Quality assessment, transparency in search strategies	(1) PRISMA compliance (2) Risk of bias (AMSTAR) (3) Automated precision and recall metrics	Fabiano et al. (2024) Shaheen et al. (2023) Siddaway et al. (2019) Susnjak et al. (2024) Zala et al. (2024)
Role of AI in Academic Literature	- Trust: Ethical AI use, human-AI collaboration - Relevance:	- Relevance: (1) Precision (2) recall of retrieval	Checco et al. (2021) Dergaa et al. (2023) Miao et al. (2024)

Summarization accuracy, retrieval precision - Traceability: Source- grounded outputs, AI usage disclosure	- Traceability: (1) Hallucination reduction (2) RAG grounding (faithfulness) - Efficiency: (1) Time savings in screening and summarization	Khalifa & Albadawy (2024) Thüs et al. (2024)
--	--	--

RAG systems offer significant advantages, such as reducing manual effort and improving the precision of academic outputs, particularly in tasks like retrieval, summarization, and content generation (Han et al., 2024). However, the core issues of trust, relevance, and traceability remain underdeveloped. Trust requires rigorous validation mechanisms, clear documentation, and transparency in AI-generated outputs (Song et al., 2024; Zhou et al., 2024). While existing metrics like faithfulness and citation accuracy provide initial benchmarks, comprehensive frameworks for evaluating bias mitigation and long-term reliability are still lacking (Roychowdhury et al., 2024).

Relevance emerges as a measurable strength of RAG systems, with metrics like precision and recall dominating the evaluation process (Sawarkar et al., 2024; Es et al., 2024). Yet, challenges persist in qualitative assessments, particularly in identifying nuanced gaps within systematic reviews (Ackerman et al., 2024). Domain-specific adaptations—such as retrieval tuning and prompt engineering—show promise in enhancing relevance but lack standard benchmarks tailored to academic literature tasks (Zhao et al., 2024; Siriwardhana et al., 2024).

Traceability, a critical factor for reproducibility and credibility, is supported by strategies like modular architecture and clear citation systems (Zhang et al., 2024; Li et al., 2024). However, gaps in ensuring reproducibility across domains remain evident, highlighting

the need for refined validation methods and traceability benchmarks (Yu et al., 2024; Venkit et al., 2024).

3 Research Method

3.1 Research Design and Method

This research employs the DSM to create a Retrieval-Augmented Generation (RAG)-based pipeline specifically tailored for systematic literature reviews. DSM is particularly suitable for developing innovative AI-driven systems because it incorporates iterative cycles of development, evaluation, and refinement informed by continuous feedback and validation (Kumar, 2024). The approach ensures the pipeline consistently aligns with practical research requirements, thus facilitating the system's adaptability and ongoing improvement based on real-world usage (Romme, 2023).

This chapter outlines the design and methodological approach adopted to develop a RAG-based pipeline tailored for conducting systematic literature reviews in the field of Technology Innovation Management (TIM). The research follows a Design Science Methodology, which emphasizes iterative development, evaluation, and refinement (Kumar, 2024; Romme, 2023). This method allowed the prototype to evolve in response to performance testing and practical implementation feedback.

The chapter begins by introducing the research design framework, followed by a step-by-step breakdown of the modular components that make up the pipeline. Each module is described in terms of its function within the system, the methods used to develop it, and its contribution to the overall literature review process. The pipeline components include research query processing, article retrieval, semantic pathway mapping, screening and quality filtering, summarization, methodology extraction, literature drafting, and evaluation.

To ensure output quality and academic rigor, an LLM judge module was also implemented. This internal evaluation component assessed generated reviews based on factual accuracy, logical coherence, and completeness (Gao et al., 2024; Song et al., 2024).

This chapter further describes the data acquisition methods used to source academic articles and how the evaluation framework was implemented. A summary table of the research method steps is provided to offer a clear overview of the activities carried out, their sequence, and the outcomes they produced.

Table 4: Research Method Steps

Step	Activity	Outcome
1	Conduct a systematic literature review to identify relevant principles, evaluation metrics, and design guidelines for RAG-based academic systems.	Set of design principles, relevant metrics, and conceptual frameworks guiding system development.
2	Design and implement a modular RAG-based pipeline, incorporating natural language processing, semantic retrieval, and summarization components.	A functional RAG system designed to streamline literature reviews
3	Create a pathway mapping module to classify queries into domain-relevant categories using semantic similarity.	A TIM pathway mapper that assigns research queries to predefined academic pathways.
4	Integrate an AI-driven query augmentation module that generates paraphrased, context-rich queries structured by the PCC framework.	A set of review queries supporting diverse retrieval contexts.
5	Test the system using multiple research queries and iteratively refine its modules based on performance feedback and implementation challenges.	A fine-tuned application with improved functionality due to testing and progressive enhancements.
6	Evaluate the system's effectiveness using predefined metrics (Relevance, Groundedness, Retrieval Relevance) and feedback from an LLM Judge.	Quantitative and qualitative evaluation results demonstrating system performance and reliability.
7	Reflect on the design process to document insights, limitations, and recommendations for future enhancement.	Insights, lessons learned , and recommendations for future iterations.

3.2 Pipeline Components

The systematic literature review pipeline follows a modular approach, clearly outlined through a structured schematic diagram (see Figure 5). This modularity supports flexibility, facilitating easy adaptation and integration of new techniques as they become available.

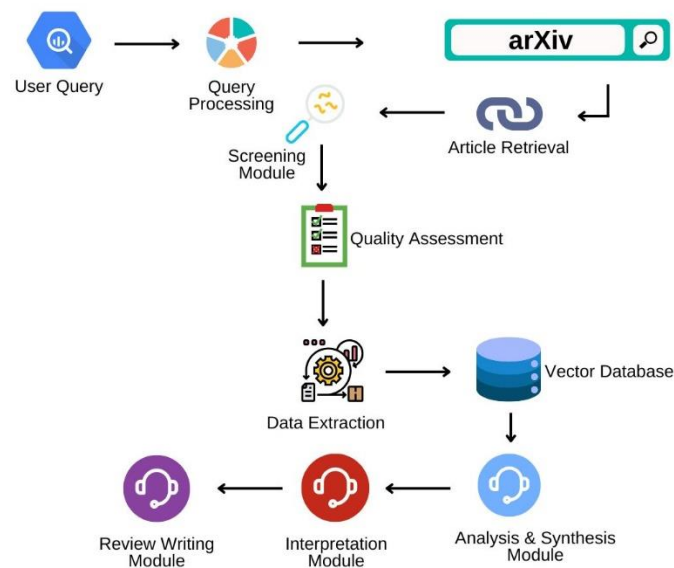


Figure 5: Schematic Diagram of the Pipeline

3.2.1 Research Query Processing

The system utilizes GPT-4 natural language processing (NLP) techniques to extract keywords and formulate research questions structured around the PCC (Population, Concept, Context) framework, widely recognized for structuring research queries effectively (Shaffril et al., 2020). These structured questions enhance the relevance and precision of article retrieval.

3.2.2 Article Retrieval

Leveraging augmented queries optimized with GPT-4, the pipeline retrieves relevant academic articles from the arXiv repository. The contextualized queries significantly improve retrieval accuracy, addressing issues of retrieval precision noted in earlier research (Zhao et al., 2024). Due to its modular nature, the pipeline supports future expansion such as integrating additional academic databases like Semantic Scholar, subject to API availability.

3.2.3 Pathway Mapping Module

To enhance retrieval precision and ensure contextual alignment with specific research interests within Technology Innovation Management (TIM), the Pathway Mapping Module (TIM Pathway Mapper) was developed and integrated into the retrieval pipeline. This module provides an automated mechanism to semantically categorize research queries.

The Technology Innovation Management program at Carleton University is dedicated to exploring emerging advancements in technology and innovation management. To enhance retrieval precision and ensure contextual alignment with specific research interests within the program, the Pathway Mapping Module (TIM Pathway Mapper) was developed and integrated into the retrieval pipeline. The research interests of this program were selected for the purposes of evaluating the proposed system's ability to categorize research queries. This module provides an automated mechanism to semantically categorize research queries, ensuring that the system's outputs align with the diverse and multidisciplinary focus areas of the program.

The module employs a pre-defined set of TIM-specific research pathways that reflect the categories of research interests of the TIM program:

- MDTE: Digital transformation and technology leadership
- MEnt: Entrepreneurship and venture creation
- MABA: Business analytics and data-driven decision-making
- MEng: Engineering research, development, and technology transfer

Each pathway is defined using specific semantic profiles encompassing target populations, core concepts, and relevant research contexts. The module utilizes semantic embeddings (SentenceTransformer: all-mpnet-base-v2) and cosine similarity measures to determine the most appropriate pathway for each query based on weighted scores considering semantic relevance, engagement significance, and temporal recency.

3.2.4 Screening & Filtering

Semantic similarity algorithms identify and remove duplicates from the retrieved articles, ensuring the uniqueness and relevance of selected papers (Roychowdhury et al., 2024). Articles are further screened using criteria such as publication date and keyword relevance, aligning closely with research objectives.

3.2.5 Quality Assessment

The pipeline assesses the quality and relevance of articles by computing semantic similarity scores between abstracts and research objectives. Only articles exceeding a predefined similarity threshold (≥ 0.5) proceed, consistent with established practices for enhancing the relevance of retrieved literature (Gao et al., 2024).

3.2.6 Summarization & Data Extraction

Extracted PDFs of qualifying articles are segmented into structured sections (abstract, introduction, methods, results, discussion) and indexed within Chroma DB, a vector database chosen for its efficiency and reliability in managing large textual datasets (Sawarkar et al., 2024).

3.2.7 Methodology Extraction and Explanation

Research methodologies are extracted from selected articles and aggregated into frequency dictionaries to identify prevalent methodological trends. Detailed explanations for the most frequent methodologies are developed, covering essential procedures, potential challenges, and applicability to specified research objectives, enhancing methodological rigor and transparency (Susnjak et al., 2024).

3.2.8 Analysis and Synthesis Module

This module systematically analyzes extracted literature data by segmenting texts into manageable sections within predefined token limits. Utilizing advanced GPT models, it synthesizes the segmented data into comprehensive narratives that identify primary research findings, prevailing methodological trends, emergent themes, and implications pertinent to the research. The synthesized outputs provide structured narratives supported by inline citations, enhancing traceability and verification of information.

Analysis & Synthesis Prompt

""You are an expert academic researcher. You have been provided with extracted data, including summaries, methods, results, conclusions, and metadata such as authors and publication details:

{chunk}

Please analyze and synthesize this data into a clear academic narrative that addresses the research question. Your narrative should include:

1. A summary of the main findings and trends.
2. Identification of key research themes.
3. Discussion of the methods employed over time in this research area.
4. A concluding synthesis of implication of findings.
5. Inline citations (e.g., Author, Year) throughout the narrative to trace factual sources.

Ensure your response is well-structured and coherent, with logical flow and clarity.

""

3.2.9 Interpretation Module

The interpretation module is responsible for providing a contextual assessment of the synthesized narratives. It explicates the significance of identified findings in relation to the research objectives, elucidates their broader implications, and pinpoints existing research gaps. Moreover, it outlines potential avenues for future research, delivering a thorough interpretative analysis that deepens the contextual relevance of synthesized insights.

Interpretation Prompt

""You are an expert academic researcher. Below is a synthesized narrative from a systematic literature review:

{input_text}

Please provide a detailed interpretation of the narrative. Address the following:

1. Explain the significance of the findings.
2. Discuss how these results contribute to achieving the research objective.
3. Identify any gaps, limitations, or areas needing further clarification.
4. Suggest future research directions based on the findings.
5. Provide inline citations (e.g., Author, Year) to support your interpretation.

Your response should be a clear, structured interpretation with distinct sections for each point.

""

3.2.10 Literature Review Writing Module

This module compiles the analytical and interpretative outcomes from preceding modules, together with methodological insights and PRISMA summaries, into coherent, structured, and academically formatted literature review drafts. Employing strategic chunking to ensure compliance with GPT token limits, it generates drafts characterized by clarity, coherence, and accurate citations. The resulting documents serve as foundational drafts for further human review and editorial refinement.

Review Writing Prompt

""You are an expert academic writer. Synthesize a concise and comprehensive literature review draft using the following inputs:

Synthesized Narrative: {inputs['synthesized_narrative']}
Interpreted Summary: {inputs['interpreted_summary']}
PRISMA Summary: {inputs['prisma_summary']}
Methodology Insights: {methodology_insights}

Ensure the review is well-structured, clearly written, and appropriately integrates these components into a cohesive narrative.

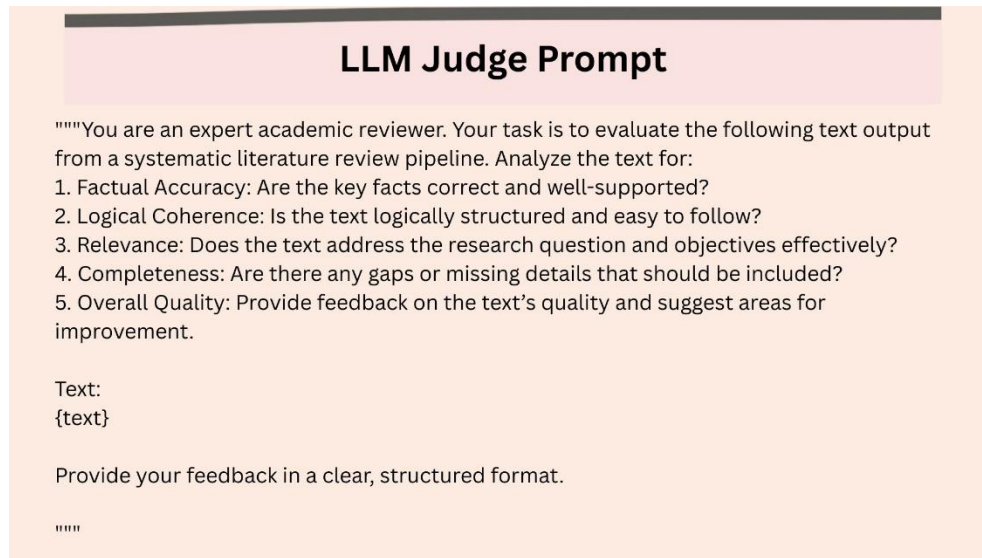
""

3.3 LLM Judge Module

The pipeline incorporates an LLM Judge Module, serving as an internal evaluation mechanism to ensure the quality and robustness of the generated literature review outputs. This module systematically assesses the final synthesized literature reviews based on critical academic criteria:

- **Factual Accuracy:** Evaluates the correctness and reliability of the factual information presented in the literature review.
- **Logical Coherence:** Checks the structured flow of ideas, ensuring clarity and logical sequencing of the narrative.
- **Relevance:** Confirms the extent to which the review addresses the stated research objectives and questions.
- **Completeness:** Detects potential informational gaps or areas that require further elaboration or clarity.

Upon completing the assessment, the LLM Judge Module provides structured, actionable feedback that outlines strengths and highlights areas for improvement, facilitating ongoing enhancement of review outputs.



3.4 Data Acquisition

Article retrieval is currently conducted via the arXiv repository. Retrieved articles are structured and indexed within Chroma DB to support efficient content retrieval and analysis.

3.5 Data Analysis

To evaluate the performance of the application three key performance metrics were evaluated—Relevance, Groundedness, and Retrieval Relevance—which were selected based on guidelines from LangChain. These metrics are critical for assessing the quality and reliability of the generated output and the effectiveness of the document retrieval component.

3.5.1 Evaluation Metrics

(i) Relevance

This metric measures the degree to which the generated literature review output aligns with the input query or research objective. It is computed using cosine similarity between the embedding of the query and the embedding of the generated text. A higher score indicates that the output addresses the query more directly.

(ii) Groundedness

This assesses how well the generated content is supported by the retrieved documents. For each retrieved document, a cosine similarity score is computed with the generated output. The aggregated score (average) serves as an indicator of the extent to which the final review is factually supported by the literature.

(iii) Retrieval Relevance

This metric evaluates the alignment of the retrieved documents with the input query. It is derived from computing the cosine similarity between the query embedding and the embedding of the retrieved documents, with the average score representing the overall relevance of the retrieved content.

(iv) Implementation Approach

The evaluation process was integrated directly into the pipeline. After executing the pipeline to generate the final literature review draft, the evaluation driver calls three separate evaluators:

- The relevance evaluator compares the query with the final review.

- The groundedness evaluator measures support by comparing the final review against the abstracts of quality-assessed articles.
- The retrieval relevance evaluator determines the overall relevance of the retrieved documents to the query.

Each run of the pipeline logs the evaluation scores, which will later be aggregated to observe trends across multiple runs.

3.6 Summary

This chapter presented the design and implementation of a modular pipeline developed to automate and streamline systematic literature reviews. Grounded in Design Science Methodology, the research adopted an iterative development approach where the system was repeatedly tested and refined using multiple research queries.

Key components of the pipeline include GPT-driven query formulation based on the PCC framework, semantic article retrieval from the arXiv repository, dynamic pathway mapping to classify research domains, article screening and quality filtering using similarity thresholds, and summarization and methodology extraction modules. These components were integrated to produce structured literature review drafts.

The system's effectiveness was evaluated using three metrics—Relevance, Groundedness, and Retrieval Relevance—and validated through an in-built LLM Judge. Quantitative scores and qualitative feedback informed ongoing refinements to the system.

Finally, the chapter documented the design steps, testing, evaluation procedures, and reflection on development outcomes. These results lay the foundation for Chapter 4,

which presents the output generated by the system and the evaluation results obtained across multiple test queries.

4 Results

This chapter presents the results obtained from the implementation and execution of the modular pipeline designed for systematic literature reviews. In alignment with the research objectives and methodology detailed in Chapters 1 and 3, the chapter is organized into distinct sections that each address a key deliverable of the study.

- Section 4.1 details the modular retrieval and summarization process. This includes the generation of the PRISMA flow diagram that documents the stages of article retrieval, screening, deduplication, and quality assessment, as well as the automated creation of a literature review draft.
- Section 4.2 describes the pathway mapping and categorization module, which employs the TIM Pathway Mapper to classify queries based on the semantic alignment of their PCC elements with predefined pathway profiles.
- Section 4.3 covers the generation of an AI-augmented query set using a GPT-based paraphrasing approach following the PCC framework.
- Section 4.4 presents the results of the methodology extraction process, highlighting the most frequently identified research methods.
- Section 4.5 offers visualizations that illustrate key outcomes, such as the temporal distribution of articles and the relationship between semantic similarity scores and publication years.
- Section 4.6 reports on the evaluation of the pipeline's performance through quantitative metrics—Relevance, Groundedness, and Retrieval Relevance—as well as qualitative assessments provided by an LLM-based judge.
- Finally, Section 4.7 provides a concise summary of the chapter.

4.1 Modular Retrieval and Summarization Pipeline

This section demonstrates the functionality of the application. The pipeline generated a literature review draft from the research query: “Transformer models for computer vision tasks” and objective "To examine the adaptation and application of transformer architectures in computer vision problems, including object detection, image classification, and semantic segmentation, with particular attention to performance comparisons with convolutional neural networks and the scalability of vision transformer approaches."

The application retrieved scholarly articles, screened and deduplicated them. A PRISMA flow diagram documents the process—from initial retrieval through screening, quality assessment, and final inclusion.

PRISMA Flow Diagram

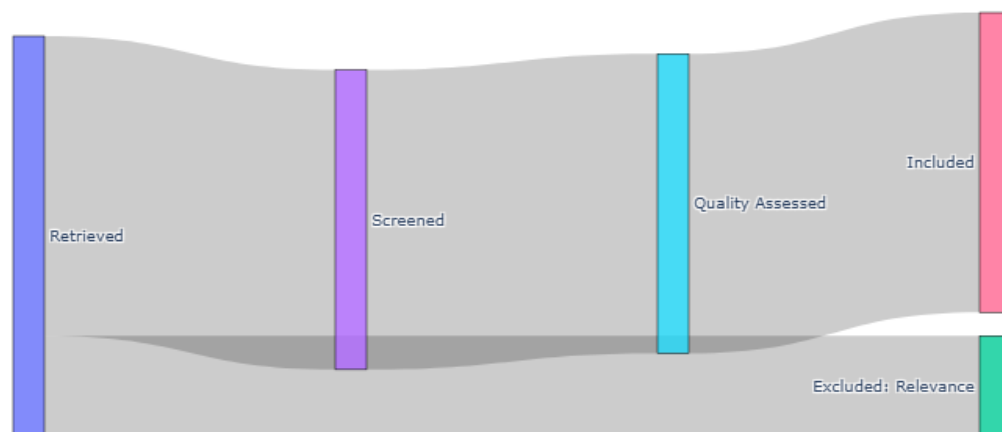


Figure 6: PRISMA Flow Diagram

The stages represented include:

- Retrieved Articles: 20
- Articles Screened: 15
- Duplicates Removed: 0
- Quality Assessed Articles: 15
- Included Articles: 15
- Excluded Articles (Relevance): 5

The field of vision-and-language tasks has seen significant advancements, particularly with the advent of Transformer-based models. However, these models often have high parameter and computation costs. Luo et al. (2022) introduced Group-wise Transformation to reduce these costs while maintaining key properties. Their lightweight Transformer, the LW-Transformer, showed competitive performance on various tasks. Fields and Kennington (2023) surveyed Vision Language Transformers, highlighting their strengths and limitations and the importance of transfer learning.

In the broader field of computer vision, Transformer-based models have shown success in tasks such as image classification, object detection, and segmentation. Pereira and Hussain (2024) emphasized the use of transfer learning with these models. The Input-Output Transformer (IO Transformer) and Output Transformer, based on SwinV2 architecture, have shown promise in evaluating the output quality of other models in computer vision tasks.

Despite their success, deploying Transformer-based models on resource-constrained edge devices has been challenging due to their compute-heavy nature. Nag et al. (2023) proposed ViTA, a hardware accelerator for inference of vision transformer models, to address this. Meanwhile, Meyer and Spruyt (2024) introduced IO Transformer and Output Transformer to evaluate the output quality of other models.

Figure 7: Literature Review Draft

The resulting literature review highlights key themes including:

- Advancements in lightweight transformer models (LW-Transformer).
- Applications of vision-language transformers in image captioning and question answering.
- Developments in hybrid vision transformers (combining convolution operations and self-attention mechanisms).
- Hardware accelerators for deploying transformer architectures efficiently.

4.2 Pathway Mapping and Categorization

The TIM Pathway Mapper categorizes queries based on the semantic similarity between the query's PCC elements and the predefined profiles for TIM pathways (MDTE, MEnt, MABA, and MEng). It computes a similarity score for each profile and assigns the query to the pathway with the highest score—provided that this score meets or exceeds a threshold of 0.5. If none of the profiles achieve a score of 0.5 or higher, the mapper returns "General," indicating that the query does not align strongly with any of the defined TIM pathways.

```
Question 1: 1. Review Question: What is the impact of transformer model architectures on the accuracy of image classification tasks?
P: Computer vision systems
C: Transformer model architectures
C: Image classification tasks
PCC Elements: {'population': 'Computer vision systems', 'concept': 'Transformer model architectures', 'context': 'Image classification tasks'}
Mapped Pathway: General

Question 2: 2. Review Question: How do transformer models compare with convolutional neural networks in terms of computational efficiency for object detection?
P: Computer vision systems
C: Transformer models versus convolutional neural networks
C: Object detection tasks
PCC Elements: {'population': 'Computer vision systems', 'concept': 'Transformer models, Convolutional neural networks', 'context': 'Object detection tasks'}
Mapped Pathway: General

Question 3: 3. Review Question: What are the challenges and limitations of implementing transformer models for real-time video analysis?
P: Real-time video analysis systems
C: Transformer models
C: Real-time processing constraints
PCC Elements: {'population': 'Real-time video analysis systems', 'concept': 'Transformer models', 'context': 'Real-time processing constraints'}
Mapped Pathway: General
```

Figure 8: Pathway Mapper Result

4.3 AI-Generated Augmented Query Set

The pipeline produced a structured set of AI-augmented queries using GPT-based paraphrasing. Queries were formulated according to the PCC (Population-Concept-Context) framework.

4.3.1 Generated Queries

The three queries generated include:

1. Question: "What is the impact of transformer model architectures on the accuracy of image classification tasks?"

- Population: Computer vision systems
 - Concept: Transformer model architectures
 - Context: Image classification tasks
2. Question: "How do transformer models compare with convolutional neural networks in terms of computational efficiency for object detection?"
- Population: Computer vision systems
 - Concept: Transformer models versus convolutional neural networks
 - Context: Object detection tasks, computational efficiency
3. Question: "What are the challenges and limitations of implementing transformer models in semantic segmentation of high-resolution images?"
- Population: High-resolution image processing
 - Concept: Transformer models
 - Context: Semantic segmentation tasks

This diversified query set enhances literature retrieval by broadening the search scope.

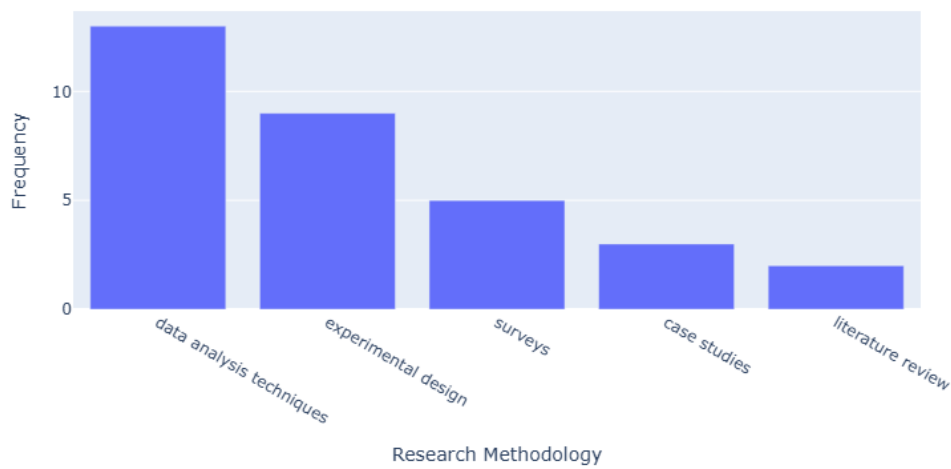
4.4 Methodology Extraction Results

The pipeline extracted and analyzed research methodologies from the retrieved articles.

The analysis identified the following top five methodologies:

- Data Analysis Techniques (13 occurrences)
- Experimental Design (8 occurrences)
- Surveys (5 occurrences)
- Case Studies (3 occurrences)
- Literature Reviews (2 occurrences)

Top 5 Research Methodologies



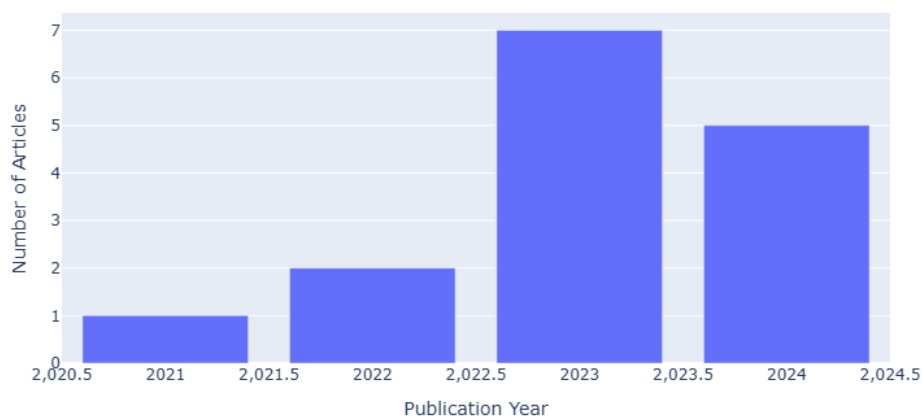
These results offer insight into the predominant research approaches in literature, which can inform subsequent analyses.

4.5 Visualization of Retrieval and Synthesis Outcomes

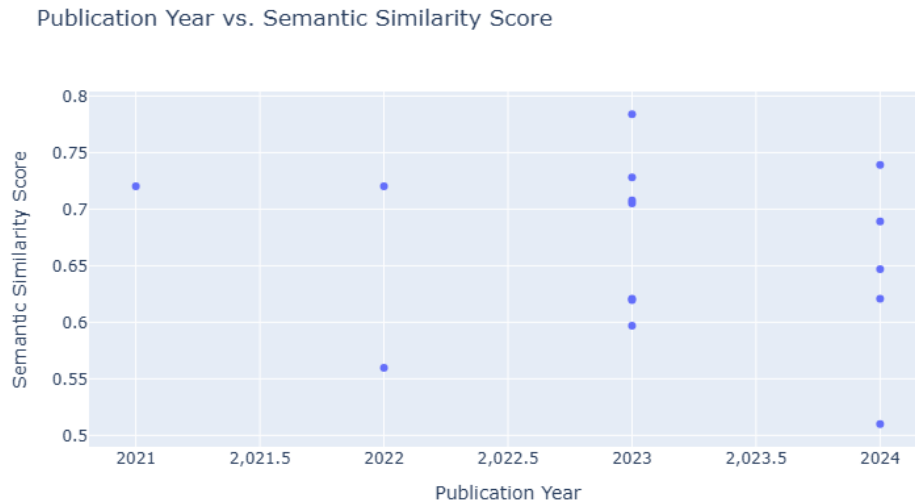
Other visual representations that provide summaries of retrieval outcomes:

- Bar Chart (Articles per Year): Showing temporal distribution of articles included in the final review.

Number of Articles per Publication Year



- Scatter Plot (Semantic Similarity vs. Publication Year): Depicting semantic similarity scores across retrieved articles.



4.6 Evaluation Results

The pipeline's performance was assessed using three quantitative metrics and qualitative LLM Judge feedback.

4.6.1 Quantitative Evaluation

The following evaluation scores were obtained from the pipeline execution:

- Relevance Score (0.55): Indicates moderate alignment between the generated review and the research query.
- Groundedness Score (0.71): Reflects strong support of the review content by the retrieved documents.
- Retrieval Relevance Score (0.51): Suggests that the retrieved documents are somewhat aligned with the query, highlighting an opportunity for improved retrieval precision.

4.6.2 Qualitative Evaluation

The LLM Judge provided structured feedback confirming the draft's factual accuracy and logical coherence, while noting incomplete citations and recommending more detailed methodological descriptions.

Feedback:

1. Factual Accuracy:

The text accurately presents key research themes and findings from the papers reviewed. The discussion on the LW-Transformer and Vision Language Transformers is well-supported with references to specific studies. The implications for future research and industrial applications are also grounded in the findings presented. Overall, the factual accuracy of the text is commendable.

2. Logical Coherence:

The text is logically structured, starting with an overview of recent advancements in vision-and-language tasks, followed by detailed discussions on specific studies, key research themes, implications for future research, and potential industrial applications. The step-by-step outlines for conducting studies using surveys, grounded theory, taxonomy methodology, and multi-scale processing methodology are presented in a clear and sequential manner. The logical flow of the text makes it easy to follow.

3. Relevance:

The text effectively addresses the research question of examining the adaptation and application of transformer architectures in computer vision problems. It provides a comprehensive overview of recent advancements in the field, key research themes, implications for future research, and potential industrial applications. The methodologies outlined for conducting studies using surveys, grounded theory, taxonomy methodology, and multi-scale processing are relevant to the research objective and offer valuable insights into the topic.

Figure 9: LLM Judge Feedback

4.7 Summary

In summary, Chapter 4 demonstrates that the integrated modular pipeline effectively retrieves, screens, synthesizes, and evaluates scholarly literature in response to user query and research objective. The results confirm that:

- (i) The retrieval and summarization processes generate a coherent literature review draft that captures key themes.
- (ii) The TIM Pathway Mapper categorizes queries by comparing PCC elements against predefined pathway profiles.

(iii) The AI-augmented query generation provides a diversified set of queries that enrich the overall literature retrieval process.

(iv) The extraction and analysis of research methodologies offer additional insights into the prevalent methods within the retrieved literature.

(v) Visual representations—including the PRISMA diagram, bar charts, and scatter plots—underscore the pipeline’s capability to organize and present complex data in an accessible format.

(vi) Quantitative evaluation metrics reveal moderate to high performance in terms of relevance and grounding, while the retrieval relevance scores suggest that further fine-tuning of the search parameters could enhance document alignment.

These outcomes validate the functionality of the application and also establish a solid foundation for the discussion of findings and potential enhancements in Chapter 5.

5 Discussion of Results

This chapter focuses on addressing the key research questions of the study, specifically exploring the effectiveness of a Retrieval-Augmented Generation (RAG) system for academic literature reviews. It discusses the findings from the data analysis, explaining their significance in relation to the research objectives. The chapter also compares the results with previous research, offering insights into the novel contributions of this project, such as the TIM Pathway Mapper and human-in-the-loop evaluation. Finally, it examines the practical implications of the findings and provides recommendations for further research and development in AI-driven literature review systems.

5.1 Project Objective and Scope

The TIM program is dedicated to exploring emerging advancements in technology. As part of this commitment, the program identified a need for more efficient, accurate, and scalable tools for conducting systematic literature reviews, aiming to enhance research processes within the academic community. Traditional literature review processes are often labor-intensive, inconsistent, and limited in scope, especially when attempting to capture emerging trends or multidisciplinary perspectives. In response, this project aimed to develop a RAG-based prototype capable of generating structured, verifiable literature review drafts with minimal manual effort. The system was designed to support end-to-end review tasks, including query processing, document retrieval, summarization, and evaluation—all tailored to the TIM research context.

5.2 Results and Research Question Alignment

The results presented in Chapter 4 show that the developed prototype successfully performs the functions outlined in the research questions introduced in Chapter 2. Specifically, the first question—concerning the ability of a RAG pipeline to automate systematic reviews—was addressed through the development and deployment of a modular system that generated literature drafts covering major themes in transformer-based computer vision research. The quality of these outputs was confirmed through the LLM Judge evaluations and supported by strong groundedness scores (~ 0.72), indicating that content was well-supported by retrieved articles.

The second question, which examined the impact of semantic query augmentation on retrieval quality, was addressed through the use of GPT-based query paraphrasing structured around the PCC framework. The generated queries improved topical coverage and diversified retrieved documents. Evaluation metrics reflected this improvement, with retrieval relevance scores (~ 0.53) and relevance scores (~ 0.51) confirming moderate alignment between queries and content.

Lastly, the third research question regarding the effectiveness of a domain-based categorization module was explored through the TIM Pathway Mapper. This module helped in contextualizing queries into one of four TIM research streams. Although some queries defaulted to a general category due to semantic ambiguity, the module generally improved the contextual precision of retrieval.

5.3 Comparison with Previous Research

Several recent studies have proposed RAG systems for academic and knowledge-intensive tasks. For instance, LitLLM (Han et al., 2024) demonstrated the effectiveness of RAG pipelines in summarizing and structuring large volumes of academic content. Similarly, Gao et al. (2024) explored hallucination reduction through grounded retrieval, and Zhao et al. (2024) emphasized the use of large language models in query contextualization. While these studies provided inspiration, the prototype developed in this project extends this work by integrating a pathway mapping module specific to the TIM domain, a structured prompting framework (PCC), and a custom evaluation method (LLM Judge).

This work advances theoretical models of RAG systems by emphasizing domain-specific customization and modular design. Most existing RAG models focus on general applications, but this project's integration of domain pathways (MDTE, MABA, MEng, MEnt) offers a more granular and tailored approach to literature retrieval. This not only refines the quality of generated content but also provides insights into the role of domain knowledge in optimizing query processing and retrieval relevance.

Furthermore, this study introduces the concept of feedback-driven iteration within RAG systems. By including a human-in-the-loop approach with the LLM Judge module and applying iterative feedback loops, this project aligns with design science principles of continuous improvement (Gregor & Hevner, 2024). This is an important theoretical contribution to the growing field of AI-driven academic tools, where the modularity and adaptability of RAG systems could be a significant advantage over traditional methods, allowing for the dynamic customization of tools based on specific academic needs.

5.4 Unexpected Findings

One of the most surprising findings was the impact of removing LangChain agents in favor of direct OpenAI API calls. Initially, LangChain agents were used for task orchestration; however, compatibility issues arose with the framework. Switching to direct API calls not only resolved these issues but unexpectedly improved the quality of generated content. Output became more detailed and structured, particularly in the methodology extraction module.

Another significant challenge was restricted access to scholarly databases. Due to the inability to secure API keys for widely used academic sources like Semantic Scholar and Elsevier, retrieval was limited to the arXiv repository. This constraint influenced the scope and representativeness of the retrieved articles, particularly affecting the coverage of research pathways such as MDTE (Digital Transformation and Technology Leadership) and MEnt (Entrepreneurship and Venture Creation). These pathways received less attention compared to MABA (Business Analytics and Data-Driven Decision Making) and MEng (Engineering Research and Technology Transfer), which were more heavily represented due to the availability of relevant literature in arXiv.

This limitation highlights the need for access to diverse and comprehensive academic sources to achieve a well-rounded literature review. It also introduces a methodological limitation: while the system functioned effectively within its operational constraints, broader coverage and more balanced representation across TIM pathways could have been achieved if these additional databases had been accessible.

The pathway mapping module occasionally failed to confidently assign a query to a TIM pathway, defaulting to 'General.' This suggests the need for better pathway definitions or

fine-tuning using a TIM-specific knowledge base. Cost constraints also restricted large-scale retrieval, forcing trade-offs in the number of articles processed per query.

5.5 Practical Implications and Lessons Learned

The insights from this project extend beyond the TIM program and have significant implications for other academic domains and industries. First, the modular design of the system, which allows for future enhancements such as integrating external scholarly APIs, could be adapted to other fields requiring literature reviews. The flexibility of the system—with its pathway mapping and semantic query augmentation—makes it applicable to a range of disciplines, from healthcare research to engineering and social sciences.

Moreover, prompt engineering, a key success factor in this project, could be generalized to other RAG-based systems in domains such as systematic reviews in healthcare or legal research, where high precision and context-specific outputs are essential. The ability to structure prompts to enhance retrieval and synthesis could be extended to automated reviews in regulatory compliance or patent research.

The findings show that even with limited infrastructure, high-quality review outputs can be achieved using targeted NLP techniques and thoughtful prompt design. This demonstrates that advanced literature review systems can be developed and deployed effectively even in resource-constrained environments, offering researchers a scalable solution without needing extensive computational power or access to vast academic repositories.

The lessons learned also suggest that future research in AI-driven systematic reviews could benefit from a more transparent and evaluable approach—integrating user

feedback into the review process to refine the system's outputs. This human-in-the-loop approach could be expanded for broader interdisciplinary applications where real-world expertise plays a critical role in validating AI-generated outputs. This project provides a foundation for future AI tools that not only reduce manual effort but also improve the quality and relevance of scholarly reviews across multiple disciplines.

Furthermore, evaluation metrics such as groundedness and relevance offer valuable insights into performance and provide a replicable framework for assessing future academic tools. This reinforces points made by Leiker et al. (2023) regarding the importance of transparent and measurable AI evaluation. By adopting these metrics, future AI-driven tools can maintain high levels of reliability and relevance, ensuring that the outputs not only meet academic standards but also align with user expectations.

From a policy perspective, the development of RAG systems for academic literature reviews could play a significant role in reshaping research funding priorities. With increasing pressure to publish quality research efficiently, funding bodies might consider supporting tools that enable faster and more accurate literature synthesis, improving the efficiency of academic workflows. Additionally, the widespread adoption of AI-driven review systems could lead to standardization in how literature reviews are conducted, potentially influencing research assessment criteria and peer review processes.

From a practice standpoint, AI tools like this could be integrated into standard research workflows in institutions, research organizations, and commercial enterprises. By enabling faster identification of research gaps, these tools could accelerate the pace of discovery in fields like innovation management, medicine, and environmental sciences, where timely and accurate literature synthesis is crucial.

5.6 Summary

This chapter provided a comprehensive discussion of the results presented in Chapter 4, reflecting on how the developed prototype aligns with the research questions and broader academic literature. The project demonstrated that the modular design of the system, along with effective use of evaluation metrics, resulted in a high-quality literature synthesis tool that successfully automates key components of systematic reviews. The prompt engineering and domain-specific pathway mapping contributed to the system's novelty, improving its accuracy and contextual relevance, which was affirmed through strong groundedness and relevance scores.

Comparisons with prior research highlighted the unique aspects of this project, such as the integration of the TIM Pathway Mapper and the adoption of a human-in-the-loop approach with the LLM Judge. These features set the project apart from similar RAG systems in academic research. The study also identified several limitations, including the restricted access to scholarly databases and occasional failures in the pathway mapping module, which impacted the coverage of certain TIM research streams.

Despite these challenges, the results underscore the potential of RAG-based systems in academic research, not only within the TIM domain but also across other disciplines. The findings suggest that such systems can be adapted to a range of academic fields, providing researchers with a scalable, reliable, and efficient tool for conducting literature reviews. The chapter also discussed practical implications, including the integration of AI tools into standard research workflows and their potential impact on policymaking in research funding and evaluation.

In conclusion, this chapter has demonstrated that, even with limited infrastructure, high-quality review outputs can be achieved through targeted NLP techniques and thoughtful prompt design. The project lays a strong foundation for future research and development in AI-driven literature review systems, offering valuable insights into enhancing the transparency and reliability of AI tools in academic contexts.

6 Conclusions, Limitations, and Future Research

Chapter 6 is organized into three sections. Section 6.1 provides the conclusion of this research. Section 6.2 identifies the limitations of the research. Section 6.3 provides suggestions for future research.

6.1 Conclusion

This project set out to develop a modular Retrieval-Augmented Generation (RAG) system tailored to the specific literature review needs of Technology Innovation Management (TIM) researchers. The goal was to design a system that could support researchers in generating reliable, traceable, and contextually relevant academic reviews using artificial intelligence.

The system developed in this project successfully integrates several components—including a semantic pathway mapping module, GPT-augmented queries based on the PCC framework, and a structured review generator—into a unified pipeline. Evaluation metrics such as relevance, groundedness, and retrieval relevance, combined with qualitative feedback from an LLM-based internal judge, confirm the system's effectiveness in producing quality outputs.

This work demonstrates that RAG systems can address key challenges in academic literature reviews, such as improving traceability, reducing hallucinations, and expanding retrieval scope through augmented queries. The project also contributes practically by offering a functional prototype and documented methodology, and theoretically by validating the use of Design Science Methodology for building AI-driven academic tools. These results support the objective set out in Chapter 1 and directly respond to the research questions outlined in Chapter 2.

In the short term, this system can serve as a valuable assistant to graduate students and researchers within the TIM program, reducing the time and effort required for literature reviews. In the long term, it offers a scalable framework for enhancing academic workflows across disciplines, especially when further integrated with external research databases and domain-specific prompting strategies.

6.2 Limitations

Despite its contributions, the project faced several limitations:

- (i) **Limited Access to Data Sources:** The application currently retrieves documents only from the arXiv repository due to API restrictions on other scholarly databases such as Semantic Scholar and Elsevier. This constrained the diversity of sources available, particularly impacting coverage in the MDTE and MEnt research pathways. Expanding data source integration in the future would enhance the system's retrieval scope and ensure broader representation across all TIM pathways.
- (ii) **Scalability Constraints:** Cost considerations restricted large-scale retrieval and analysis. The system processed a relatively small set of articles per query due to the high token cost of current LLMs with smaller context windows. This can be mitigated in future iterations by adopting more affordable models with larger context windows and implementing chunking strategies to process larger datasets efficiently. These improvements would enhance scalability without compromising performance or cost-effectiveness.
- (iii) **Pathway Classification Challenges:** Although the pathway mapping module generally improved query contextualization, some queries did not clearly align

with the predefined TIM categories and defaulted to the “General” category.

This suggests a need for more refined pathway definitions or training data tailored to TIM research themes. Further development is required to improve the precision of query categorization.

6.3 Suggestions for Future Research

Future projects can build upon this work by addressing the identified limitations in the following ways:

- (i) **Broaden Data Source Integration:** Efforts should be made to integrate APIs from diverse scholarly repositories such as Semantic Scholar, Scopus, and Elsevier to improve retrieval breadth and representation across all TIM pathways.
- (ii) **Enhance Pathway Mapping Accuracy:** Future research should explore refining the pathway mapping module by training it on TIM-specific datasets or incorporating supervised learning techniques to improve semantic alignment.
- (iii) **Explore Prompt Engineering Strategies:** The project revealed that prompt engineering significantly influences system performance. Future studies could investigate how prompt design can be optimized for specific academic domains by establishing best practices and exploring the adaptability of prompting techniques across different research disciplines.
- (iv) **Evaluate with Broader Queries:** Additional testing with a wider range of research queries would help validate the system’s consistency and adaptability. Comparative studies using queries from different TIM pathways can reveal potential biases or weaknesses in the retrieval process.

- (v) **Develop Interactive User Interfaces:** Although a basic user interface was implemented, future development could focus on creating a fully interactive interface where users can refine queries, adjust parameters, and provide feedback in real-time. This would improve user experience and facilitate iterative improvements to the system.
- (vi) **Explore Additional RAG Features:** A list of 28 potential features for RAG-based systems has been included in the appendix. Future researchers are encouraged to review this feature set and evaluate which components can be implemented or enhanced in subsequent studies. This exploration could guide future development and contribute to building feature-rich academic support systems.

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Appendix A: Features of a RAG LLM

Feature	Category	Description	Relevance
Distractor Document Handling	Retrieval Optimization	Differentiates between relevant and irrelevant documents.	Improves RAG accuracy by excluding irrelevant content.
Faithfulness and Factual Correctness Metrics	Metric Improvement	Uses metrics like RAGAS to evaluate truthfulness and source alignment.	Enhances trust and traceability of generated content.
Domain-Adaptive Embedding Models	Retrieval Model Tuning	Embedding models fine-tuned for specific domains.	Improves domain-specific retrieval performance.
Chain-of-Thought Reasoning	Generation Optimization	Injects logical reasoning steps into generative output.	Improves answer coherence and contextual accuracy.
Intermediate Output Visibility	Evaluation Techniques	Makes metric computations transparent.	Builds user trust and enables easier system auditing.
Instruction Fine-Tuning	Domain Specialization	Trains LLMs on domain-specific queries.	Increases accuracy for literature reviews in specialized fields.
Enhanced Relevance Filtering	Filtering Techniques	Prioritizes relevant content during retrieval.	Boosts precision of retrieved context.
Hybrid Query-Based Retrieval	Retrieval Optimization	Combines dense and sparse retrieval methods.	Improves precision and recall through hybrid indexing.
Domain-Specific Prompt Tuning	Generation Optimization	Adapts prompts to improve contextual alignment.	Enhances relevance in output.
Factuality Metrics (e.g., TruthfulQA)	Metrics	Measures factual consistency of outputs.	Ensures reliability and minimizes hallucinations.
Interactive Retrieval Techniques	Retrieval Optimization	Enables user interaction to refine search queries.	Improves retrieval relevance and transparency.
Semantic Filtering Algorithms	Filtering Techniques	Filters content using semantic matching.	Reduces noise, increases relevance.
Knowledge Augmentation	Data Handling	Adds information from external knowledge bases.	Enhances factual accuracy and contextual depth.
Faithfulness and Transparency	Metrics	Ensures outputs are aligned with citations and reduce hallucination.	Builds traceability and trust.
Balanced Information Presentation	Generation Optimization	Avoids biased, one-sided responses.	Encourages diversity and reliability in content.
Interactivity and Context Awareness	User Interface Features	Enables query clarification and source validation.	Enhances usability and user trust.
Efficiency in Retrieval and Summarization	Retrieval Optimization	Avoids redundant documents and	Increases relevance and completeness.

		improves summarization quality.	
Metrics for Evaluation	Evaluation Techniques	Includes accuracy, relevance, faithfulness, etc.	Provides a framework for assessing system reliability.
Explicit Failure Point Analysis	Evaluation Methods	Identifies recurring errors in RAG systems.	Guides improvement of system architecture and robustness.
Adaptive Retrieval Mechanisms	Retrieval Systems	Dynamically adjusts retrieval based on input or context.	Enhances contextual adaptation and output relevance.
Multilayered Traceability Framework	Data Handling	Aligns outputs with source credibility through layered validation.	Strengthens traceability and quality control.
Robust Error Categorization	Evaluation Methods	Classifies errors to guide improvements.	Streamlines debugging and system refinement.
Dynamic Embedding Models	Retrieval	Embedding models adapt based on query or domain.	Enables domain-specific accuracy.
Fine-Grained Segmentation	Data Processing	Breaks documents into smaller, meaningful parts.	Improves precision of retrieval.
Reranking with Relevance Metrics	Metrics	Reorders result by relevance to the query.	Boosts answer quality and prioritization.
Sentence-Based Planning	Generation	Structures output with predefined sentence templates.	Enhances coherence and factual accuracy.
Query Augmentation	Retrieval	Expands queries to capture diverse ideas.	Encourages broader coverage and deeper insight.
Modular RAG Framework	Architecture	Separates system components for flexibility.	Supports system scalability, adaptability, and modular testing.

Appendix B: Application Source Code

Double-click the icon below to open the source code (PDF).

It is also accessible via the GitHub repository:

https://github.com/abibatoki/Prompt-Engineering/blob/systematic-review/application_code.py



application_code.pdf

Appendix C: Application Interface

Double-click the icon below to open the application interface (PDF).



Application Interface.pdf

Systematic Literature Review

This interface runs the systematic review pipeline and displays the final review, judge feedback, PRISMA summary, visualizations, and evaluation results.

Query Enter your search query here Research Objective Enter the research objective Start Year 2020 Article Limit 10 20 100 Submit Clear	Final Literature Review Draft LLM Judge Feedback PRISMA Summary
---	--

Appendix D: Dataset of Retrieved Papers (Rayyan Import CSV)

Title	Abstract	Year	Link	Source
Towards Lightweight Transformer via Group-wise Transformation for...	Despite the exciting performance, Transformer is criticized for its...	2022	Link	arXiv
Vision Language Transformers: A Survey	Vision language tasks, such as answering questions about or...	2023	Link	arXiv
A Review of Transformer-Based Models for Computer Vision Tasks...	Transformer-based models have transformed the landscape of...	2023	Link	arXiv
IO Transformer: Evaluating SwinV2-Based Reward Models for...	Transformers and their derivatives have achieved state-of-the-art...	2023	Link	arXiv
ViTA: A Vision Transformer Inference Accelerator for Edge...	Vision Transformer models, such as ViT, Swin Transformer, and...	2023	Link	arXiv

Note: This table shows a preview. The full CSV is included as a separate file and embedded below.



rayyan_import_retrieved_articles.csv



Dear Abibat,

Thank you for your interest in a Semantic Scholar API key! We have received your request. **Please note that our API key issuance is currently on pause due to high demand.** We'll resume the process as soon as keys become available again. Thank you for your patience!

In the interim, most endpoints are available to you as an unauthenticated user, with a lower rate limit. We encourage you to review our [tutorial](#) to familiarize yourself with Semantic Scholar API.

Thank you,
Semantic Scholar team